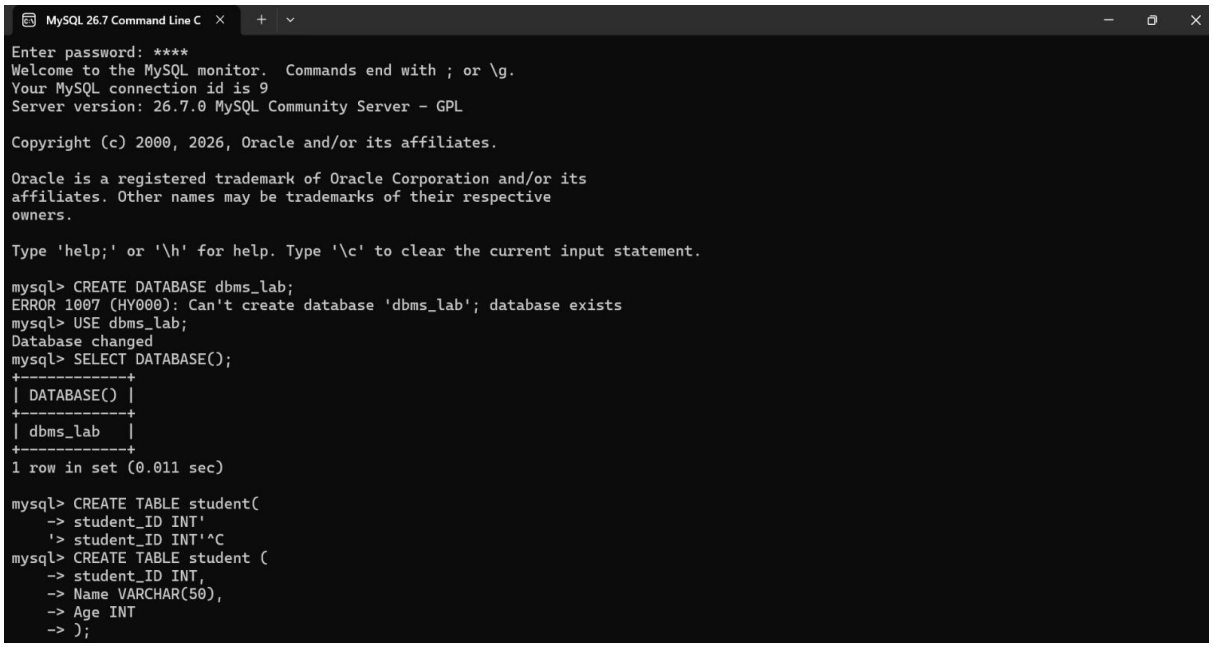
1)DDL Commands using CREATE, ALTER , TRUNCATE and DROP

Output



```
MySQL 26.7 Command Line C

Enter password: ****
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 26.7.0 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE dbms_lab;
ERROR 1007 (HY000): Can't create database 'dbms_lab'; database exists
mysql> USE dbms_lab;
Database changed
mysql> SELECT DATABASE();
+------------+
| DATABASE() |
+------------+
| dbms_lab   |
+------------+
1 row in set (0.011 sec)

mysql> CREATE TABLE student(
    -> student_ID INT'
    '> student_ID INT'^C
mysql> CREATE TABLE student (
    -> student_ID INT,
    -> Name VARCHAR(50),
    -> Age INT
    -> );
```



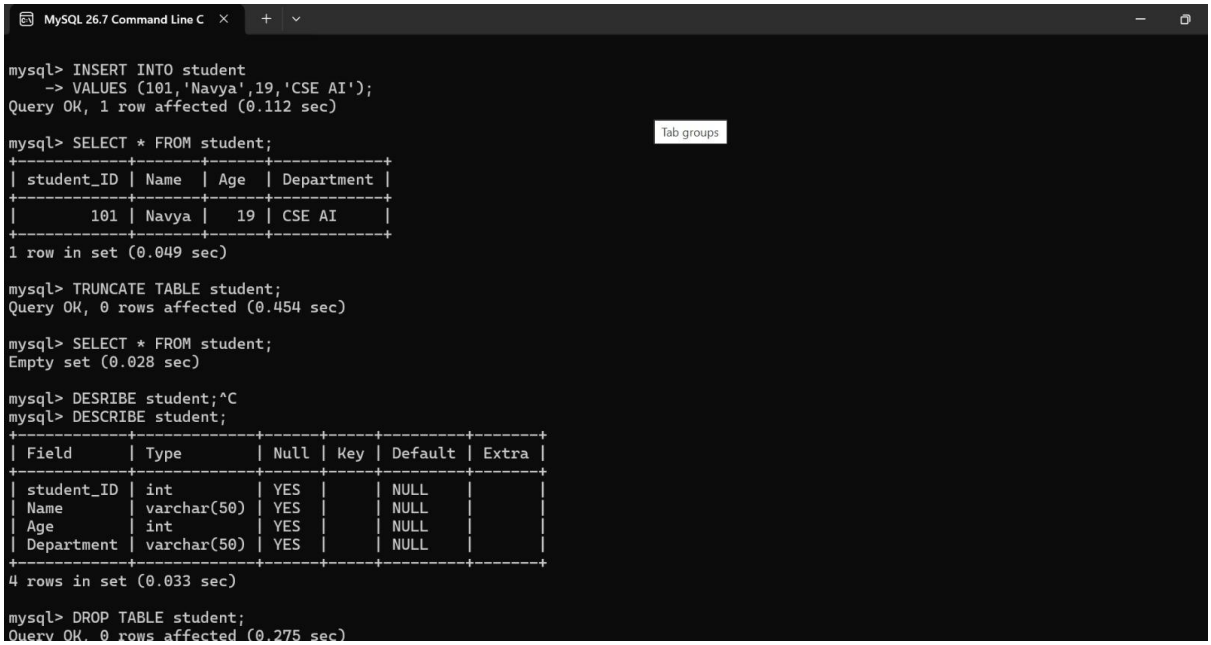
```
MySQL 26.7 Command Line C

    -> );
Query OK, 0 rows affected (1.297 sec)

mysql> DESCRIBE student;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| student_ID | int         | YES  |     | NULL    |       |
| Name       | varchar(50) | YES  |     | NULL    |       |
| Age        | int         | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
3 rows in set (0.491 sec)

mysql> ALTER TABLE student
    -> ADD Department VARACHAR(50);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the righ
t syntax to use near 'VARACHAR(50)' at line 2
mysql> ALTER TABLE student
    -> ADD Department VARCHAR(50);
Query OK, 0 rows affected (0.797 sec)
Records: 0  Duplicates: 0  Warnings: 0

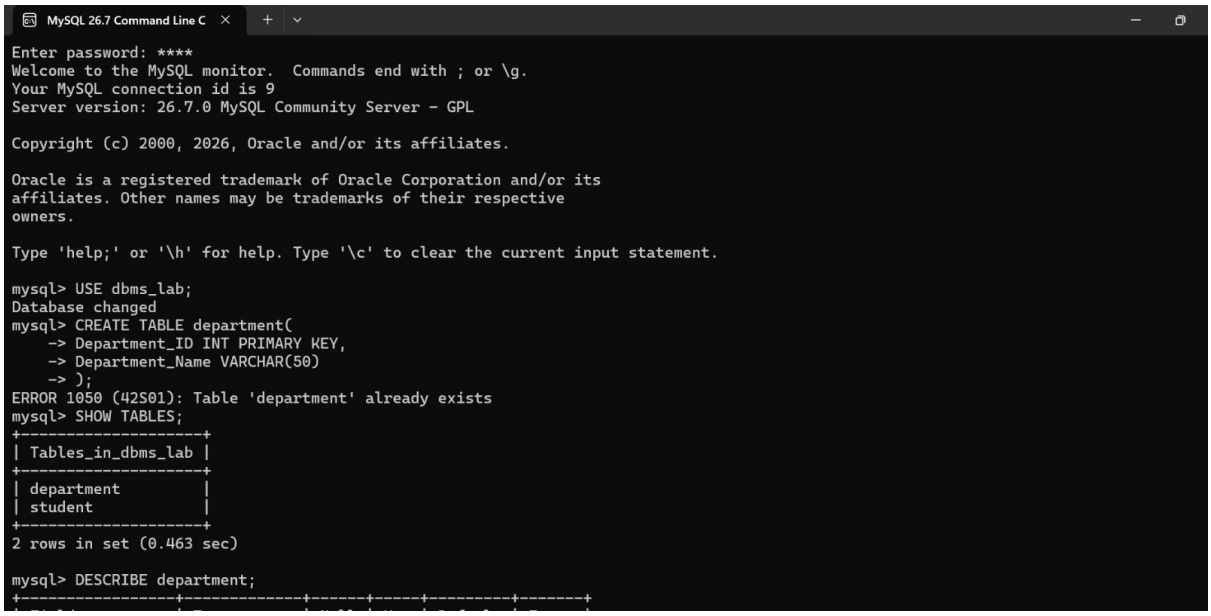
mysql> DESCRIBE student;
+------------+-------------+------+-----+---------+-------+
| Field      | Type        | Null | Key | Default | Extra |
+------------+-------------+------+-----+---------+-------+
| student_ID | int         | YES  |     | NULL    |       |
| Name       | varchar(50) | YES  |     | NULL    |       |
| Age        | int         | YES  |     | NULL    |       |
| Department | varchar(50) | YES  |     | NULL    |       |
+------------+-------------+------+-----+---------+-------+
4 rows in set (0.034 sec)
```

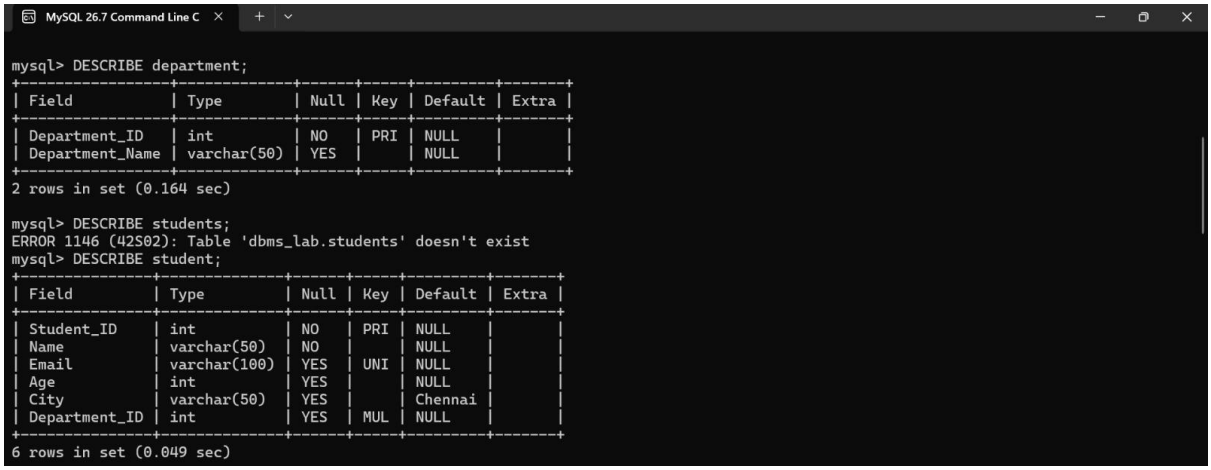


2) DDL Commands with Constraints – PRIMARY, FOREIGN KEY, UNIQUE,

CHECK, DEFAULT, NOT NULL

Output:





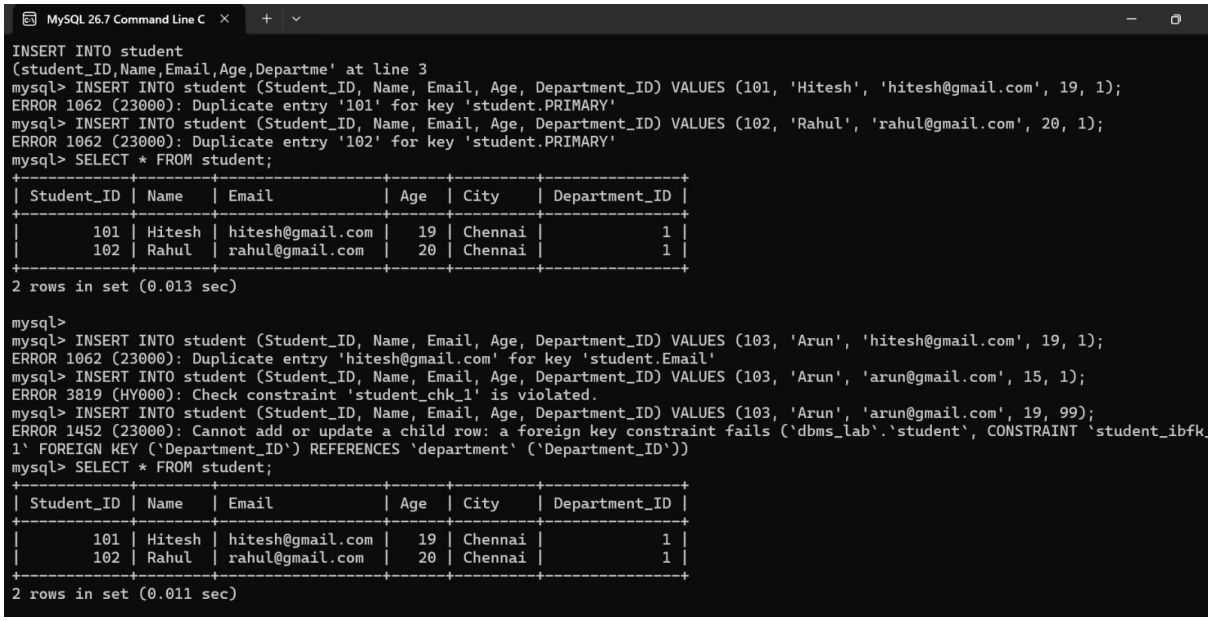

```
mysql> DESCRIBE department;
+-----------------+-------------+------+-----+---------+-------+
| Field           | Type        | Null | Key | Default | Extra |
+-----------------+-------------+------+-----+---------+-------+
| Department_ID   | int         | NO   | PRI | NULL    |       |
| Department_Name | varchar(50) | YES  |     | NULL    |       |
+-----------------+-------------+------+-----+---------+-------+
2 rows in set (0.164 sec)

mysql> DESCRIBE students;
ERROR 1146 (42S02): Table 'dbms_lab.students' doesn't exist
mysql> DESCRIBE student;
+---------------+--------------+------+-----+---------+-------+
| Field         | Type         | Null | Key | Default | Extra |
+---------------+--------------+------+-----+---------+-------+
| Student_ID    | int          | NO   | PRI | NULL    |       |
| Name          | varchar(50)  | NO   |     | NULL    |       |
| Email         | varchar(100) | YES  | UNI | NULL    |       |
| Age           | int          | YES  |     | NULL    |       |
| City          | varchar(50)  | YES  |     | Chennai |       |
| Department_ID | int          | YES  | MUL | NULL    |       |
+---------------+--------------+------+-----+---------+-------+
6 rows in set (0.049 sec)
```

```
mysql> SHOW CREATE TABLE student\G
*************************** 1. row ***************************
       Table: student
Create Table: CREATE TABLE `student` (
  `Student_ID` int NOT NULL,
  `Name` varchar(50) NOT NULL,
  `Email` varchar(100) DEFAULT NULL,
  `Age` int DEFAULT NULL,
  `City` varchar(50) DEFAULT 'Chennai',
  `Department_ID` int DEFAULT NULL,
  PRIMARY KEY (`Student_ID`),
  UNIQUE KEY `Email` (`Email`),
  KEY `Department_ID` (`Department_ID`),
  CONSTRAINT `student_ibfk_1` FOREIGN KEY (`Department_ID`) REFERENCES `department` (`Department_ID`),
  CONSTRAINT `student_chk_1` CHECK ((`Age` >= 17))
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_0900_ai_ci
1 row in set (0.019 sec)

mysql> SELECT * FROM department;
+---------------+-----------------+
| Department_ID | Department_Name |
+---------------+-----------------+
|             1 | CSE             |
|             2 | ECE             |
|             3 | MECH            |
+---------------+-----------------+
3 rows in set (0.216 sec)

mysql> INSERT INTO student
    -> (student_ID,Name,Email,Age,Department_ID)
    -> VALUES(101,'Navya',navya@gmail.com',19,1);
    '>
```

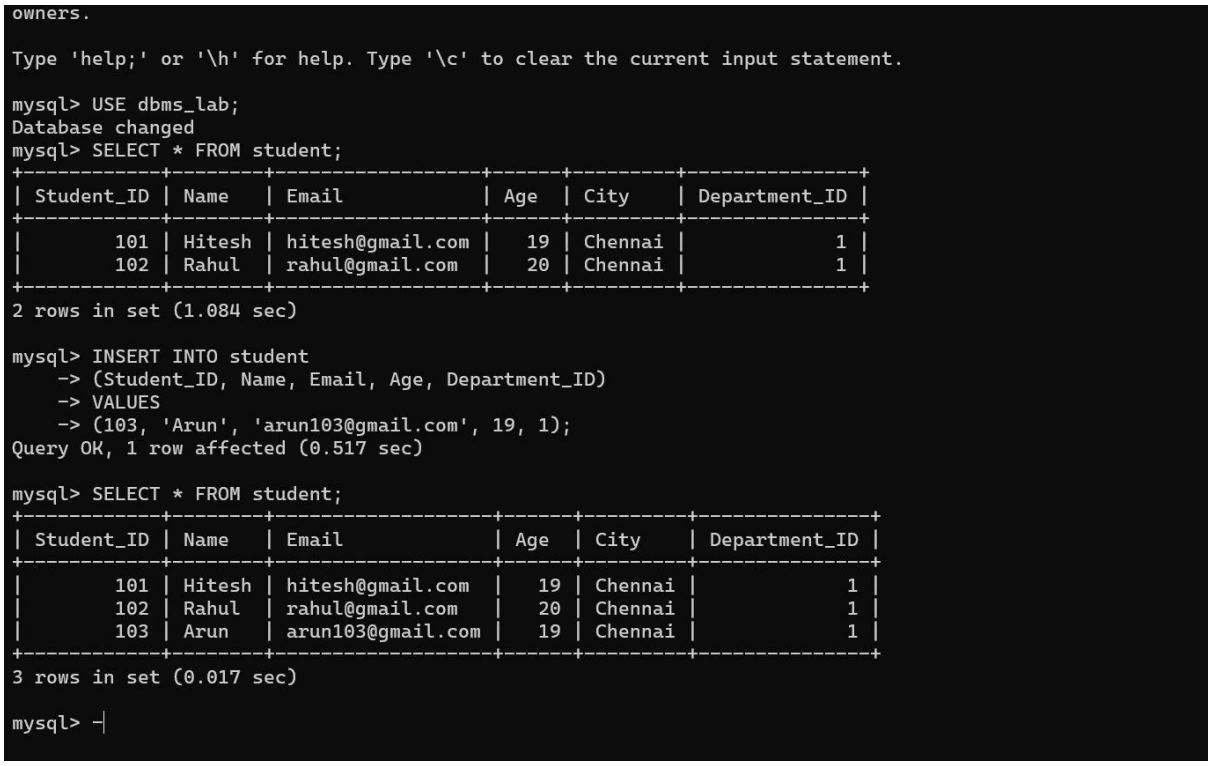



```
INSERT INTO student
(student_ID,Name,Email,Age,Departme' at line 3
mysql> INSERT INTO student (Student_ID, Name, Email, Age, Department_ID) VALUES (101, 'Hitesh', 'hitesh@gmail.com', 19, 1);
ERROR 1062 (23000): Duplicate entry '101' for key 'student.PRIMARY'
mysql> INSERT INTO student (Student_ID, Name, Email, Age, Department_ID) VALUES (102, 'Rahul', 'rahul@gmail.com', 20, 1);
ERROR 1062 (23000): Duplicate entry '102' for key 'student.PRIMARY'
mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.013 sec)

mysql>
mysql> INSERT INTO student (Student_ID, Name, Email, Age, Department_ID) VALUES (103, 'Arun', 'hitesh@gmail.com', 19, 1);
ERROR 1062 (23000): Duplicate entry 'hitesh@gmail.com' for key 'student.Email'
mysql> INSERT INTO student (Student_ID, Name, Email, Age, Department_ID) VALUES (103, 'Arun', 'arun@gmail.com', 15, 1);
ERROR 3819 (HY000): Check constraint 'student_chk_1' is violated.
mysql> INSERT INTO student (Student_ID, Name, Email, Age, Department_ID) VALUES (103, 'Arun', 'arun@gmail.com', 19, 99);
ERROR 1452 (23000): Cannot add or update a child row: a foreign key constraint fails (`dbms_lab`.`student`, CONSTRAINT `student_ibfk_
1` FOREIGN KEY (`Department_ID`) REFERENCES `department` (`Department_ID`))
mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.011 sec)
```

3)DML Commands using INSERT, SELECT

Output:



```
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+-------------------+------+---------+---------------+
| Student_ID | Name   | Email             | Age  | City    | Department_ID |
+------------+--------+-------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com  |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com   |   20 | Chennai |             1 |
+------------+--------+-------------------+------+---------+---------------+
2 rows in set (1.084 sec)

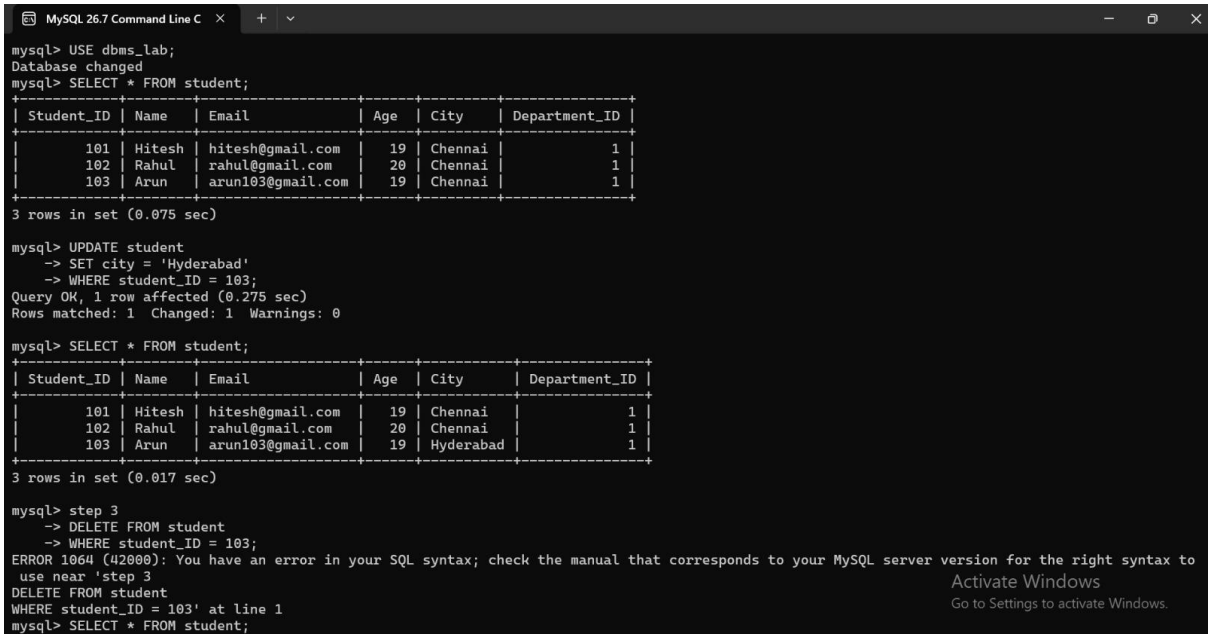
mysql> INSERT INTO student
    -> (Student_ID, Name, Email, Age, Department_ID)
    -> VALUES
    -> (103, 'Arun', 'arun103@gmail.com', 19, 1);
Query OK, 1 row affected (0.517 sec)

mysql> SELECT * FROM student;
+------------+--------+-------------------+------+---------+---------------+
| Student_ID | Name   | Email             | Age  | City    | Department_ID |
+------------+--------+-------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com  |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com   |   20 | Chennai |             1 |
|        103 | Arun   | arun103@gmail.com |   19 | Chennai |             1 |
+------------+--------+-------------------+------+---------+---------------+
3 rows in set (0.017 sec)

mysql>
```

4) DML Commands using UPDATE, DELETE

Output:



```
MySQL 26.7 Command Line C

mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+-------------------+------+---------+---------------+
| Student_ID | Name   | Email             | Age  | City    | Department_ID |
+------------+--------+-------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com  |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com   |   20 | Chennai |             1 |
|        103 | Arun   | arun103@gmail.com |   19 | Chennai |             1 |
+------------+--------+-------------------+------+---------+---------------+
3 rows in set (0.075 sec)

mysql> UPDATE student
    -> SET city = 'Hyderabad'
    -> WHERE student_ID = 103;
Query OK, 1 row affected (0.275 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student;
+------------+--------+-------------------+------+-----------+---------------+
| Student_ID | Name   | Email             | Age  | City      | Department_ID |
+------------+--------+-------------------+------+-----------+---------------+
|        101 | Hitesh | hitesh@gmail.com  |   19 | Chennai   |             1 |
|        102 | Rahul  | rahul@gmail.com   |   20 | Chennai   |             1 |
|        103 | Arun   | arun103@gmail.com |   19 | Hyderabad |             1 |
+------------+--------+-------------------+------+-----------+---------------+
3 rows in set (0.017 sec)

mysql> step 3
    -> DELETE FROM student
    -> WHERE student_ID = 103;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
 use near 'step 3
DELETE FROM student
WHERE student_ID = 103' at line 1
mysql> SELECT * FROM student;
```

Activate Windows
Go to Settings to activate Windows.

```
+------------+--------+-------------------+------+-----------+---------------+
3 rows in set (0.017 sec)

mysql> step 3
    -> DELETE FROM student
    -> WHERE student_ID = 103;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to
 use near 'step 3
DELETE FROM student
WHERE student_ID = 103' at line 1
mysql> SELECT * FROM student;
+------------+--------+-------------------+------+-----------+---------------+
| Student_ID | Name   | Email             | Age  | City      | Department_ID |
+------------+--------+-------------------+------+-----------+---------------+
|        101 | Hitesh | hitesh@gmail.com  |   19 | Chennai   |             1 |
|        102 | Rahul  | rahul@gmail.com   |   20 | Chennai   |             1 |
|        103 | Arun   | arun103@gmail.com |   19 | Hyderabad |             1 |
+------------+--------+-------------------+------+-----------+---------------+
3 rows in set (0.013 sec)

mysql> DELETE FROM student
    -> WHERE Student_ID = 103;
Query OK, 1 row affected (0.116 sec)

mysql> SELECT * FROM student;
+------------+--------+-----------------+------+---------+---------------+
| Student_ID | Name   | Email           | Age  | City    | Department_ID |
+------------+--------+-----------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+-----------------+------+---------+---------------+
2 rows in set (0.012 sec)

mysql>
```

Activate Windows
Go to Settings to activate Windows.

5)TCL Commands – COMMIT, SAVEPOINT, ROLLBACK

Output:

```
mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+-----------------+------+---------+---------------+
| Student_ID | Name   | Email           | Age  | City    | Department_ID |
+------------+--------+-----------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+-----------------+------+---------+---------------+
2 rows in set (0.021 sec)

mysql> START TRANSACTION;
Query OK, 0 rows affected (0.060 sec)

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.055 sec)

mysql> UPDATE student
    -> SET City = 'Bangalore'
    -> WHERE Student_ID = 101;
Query OK, 1 row affected (0.041 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM student;
+------------+--------+-----------------+------+-----------+---------------+
| Student_ID | Name   | Email           | Age  | City      | Department_ID |
+------------+--------+-----------------+------+-----------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Bangalore |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai   |             1 |
+------------+--------+-----------------+------+-----------+---------------+
2 rows in set (0.021 sec)

mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.079 sec)
```

```
mysql> SELECT * FROM student;
+------------+--------+------------------+------+-----------+---------------+
| Student_ID | Name   | Email            | Age  | City      | Department_ID |
+------------+--------+------------------+------+-----------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Bangalore |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai   |             1 |
+------------+--------+------------------+------+-----------+---------------+
2 rows in set (0.021 sec)

mysql> ROLLBACK TO SAVEPOINT sp1;
Query OK, 0 rows affected (0.079 sec)

mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.015 sec)

mysql>
```

6. DCL Commands – GRANT, REVOKE

Output:

```
MySQL 26.7 Command Line C

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE dbms_lab;
Database changed
mysql> CREATE USER 'student1'@'localhost' IDENTIFIED BY'student@123;
    '>
    '> ^C
mysql> CREATE USER 'student1'@'localhost' IDENTIFIED BY 'Student@123';
Query OK, 0 rows affected (0.642 sec)

mysql> REVOKE SELECT ON dbms_lab.student FROM 'student1'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'student1' on host 'localhost' on table 'student'
mysql> REVOKE SELECT ON dbms_lab.student FROM 'student1'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'student1' on host 'localhost' on table 'student'
mysql> GRANT SELECT ON dbms_lab.student TO 'student1'@'localhost';
Query OK, 0 rows affected (0.086 sec)

mysql> REVOKE SELECT ON dbms_lab.student FROM 'student1'@'localhost';
Query OK, 0 rows affected (0.086 sec)

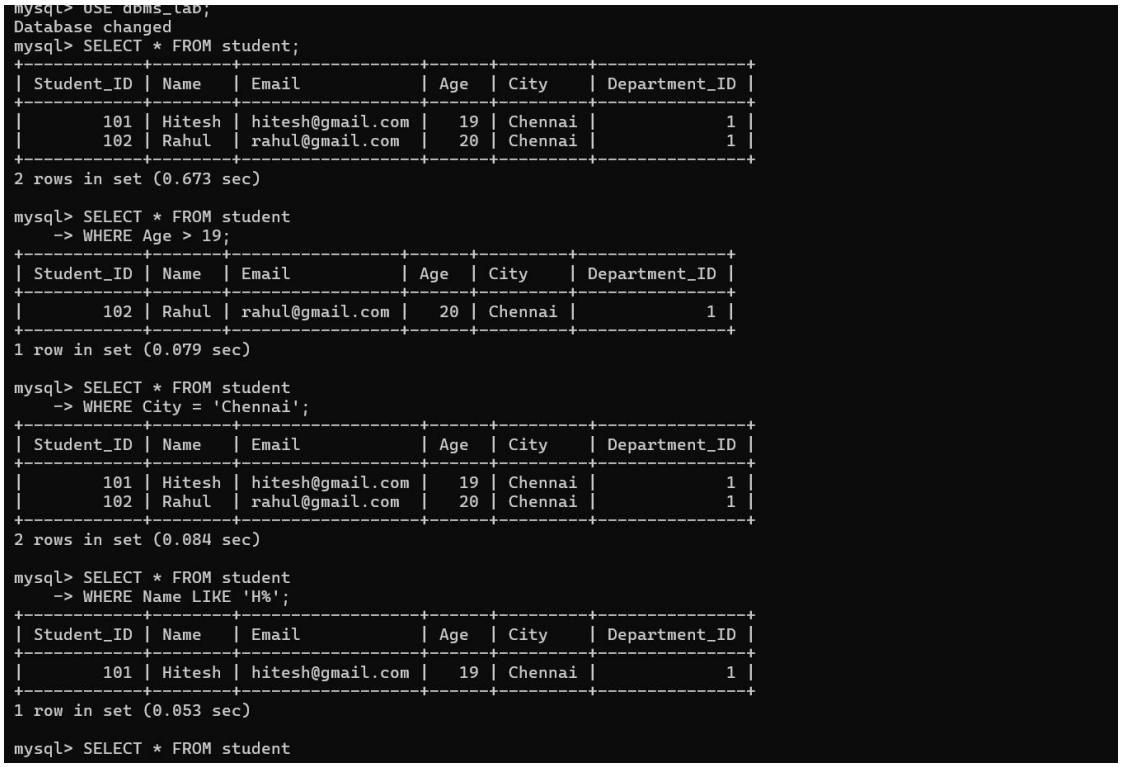
mysql> SHOW GRANTS FOR 'student1'@'localhost';
+-----------------------------------------------+
| Grants for student1@localhost                 |
+-----------------------------------------------+
| GRANT USAGE ON *.* TO `student1`@`localhost` |
+-----------------------------------------------+
1 row in set (0.097 sec)

mysql> SHOW GRANTS FOR 'student1'@'localhost';
+-----------------------------------------------+
| Grants for student1@localhost                 |
+-----------------------------------------------+
| GRANT USAGE ON *.* TO `student1`@`localhost` |
+-----------------------------------------------+
1 row in set (0.019 sec)

mysql>
```

7. SELECT with various clauses – WHERE, pattern matching

Output:



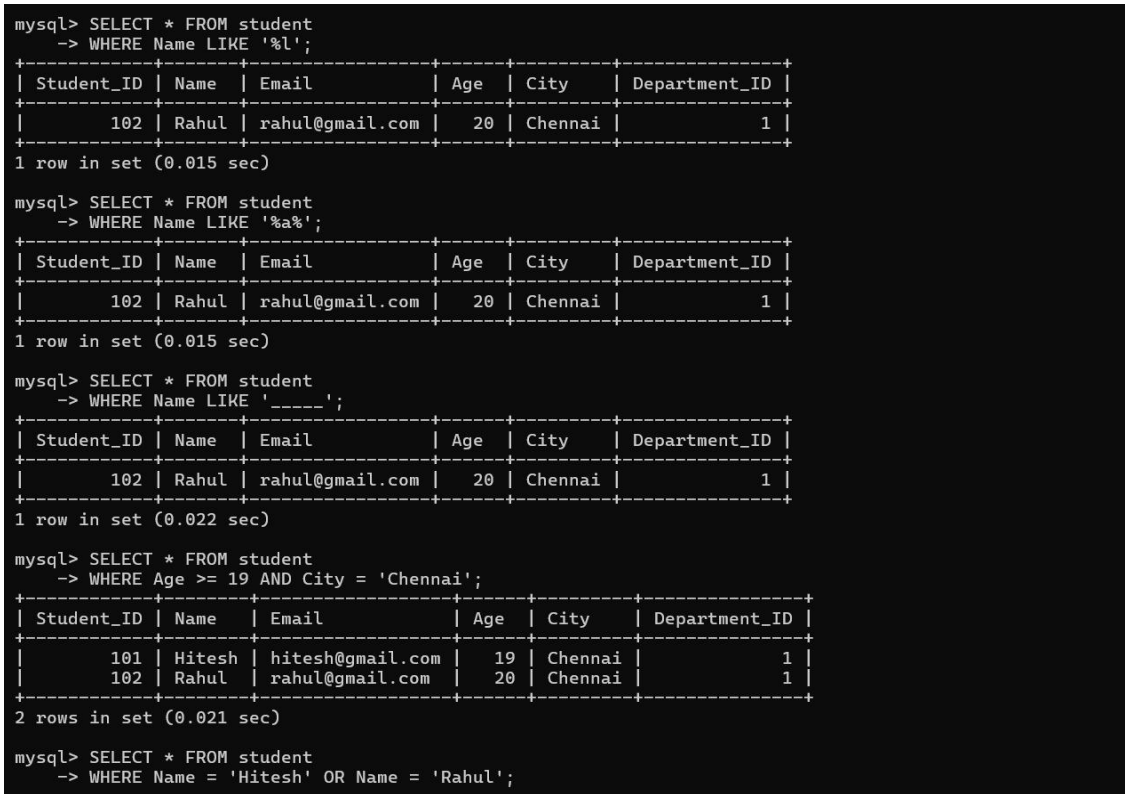
```
mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.673 sec)

mysql> SELECT * FROM student
    -> WHERE Age > 19;
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.079 sec)

mysql> SELECT * FROM student
    -> WHERE City = 'Chennai';
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.084 sec)

mysql> SELECT * FROM student
    -> WHERE Name LIKE 'H%';
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
1 row in set (0.053 sec)

mysql> SELECT * FROM student
```



```
mysql> SELECT * FROM student
    -> WHERE Name LIKE '%l';
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.015 sec)

mysql> SELECT * FROM student
    -> WHERE Name LIKE '%a%';
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.015 sec)

mysql> SELECT * FROM student
    -> WHERE Name LIKE '_____';
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.022 sec)

mysql> SELECT * FROM student
    -> WHERE Age >= 19 AND City = 'Chennai';
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.021 sec)

mysql> SELECT * FROM student
    -> WHERE Name = 'Hitesh' OR Name = 'Rahul';
```

```
    -> WHERE Name = 'Hitesh' OR Name = 'Rahul';
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.017 sec)

mysql> SELECT * FROM student
    -> WHERE City <> 'Chennai';
Empty set (0.015 sec)

mysql> SELECT * FROM student
    -> WHERE Name LIKE 'R%';
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.021 sec)

mysql> SELECT * FROM student
    -> WHERE Name LIKE '%it%';
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
1 row in set (0.017 sec)

mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.011 sec)
```

8. SELECT with various clauses – BETWEEN, IN, Aggregate Function

Output:

```
mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.016 sec)

mysql> SELECT * FROM student
    -> WHERE Age BETWEEN 19 AND 20;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.058 sec)

mysql> SELECT * FROM student
    -> WHERE City IN ('Chennai', 'Hyderabad');
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.053 sec)

mysql> SELECT * FROM student
    -> WHERE Name NOT IN ('Hitesh');
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.045 sec)
```

```
1 row in set (0.045 sec)

mysql> SELECT COUNT(*) AS Total_Students
    -> FROM student;
+----------------+
| Total_Students |
+----------------+
|              2 |
+----------------+
1 row in set (0.077 sec)

mysql> SELECT SUM(Age) AS Total_Age
    -> FROM student;
+-----------+
| Total_Age |
+-----------+
|        39 |
+-----------+
1 row in set (0.055 sec)

mysql> SELECT AVG(Age) AS Average_Age
    -> FROM student;
+-------------+
| Average_Age |
+-------------+
|     19.5000 |
+-------------+
1 row in set (0.050 sec)

mysql> SELECT MAX(Age) AS Maximum_Age
    -> FROM student;
+-------------+
| Maximum_Age |
+-------------+
|          20 |
+-------------+
1 row in set (0.021 sec)

mysql> SELECT MIN(Age) AS Minimum_Age
    -> FROM student;
```

```
1 row in set (0.021 sec)

mysql> SELECT MIN(Age) AS Minimum_Age
    -> FROM student;
+-------------+
| Minimum_Age |
+-------------+
|          19 |
+-------------+
1 row in set (0.060 sec)

mysql> SELECT
    ->     COUNT(*) AS Total_Students,
    ->     SUM(Age) AS Total_Age,
    ->     AVG(Age) AS Average_Age,
    ->     MAX(Age) AS Maximum_Age,
    ->     MIN(Age) AS Minimum_Age
    -> FROM student;
+----------------+-----------+-------------+-------------+-------------+
| Total_Students | Total_Age | Average_Age | Maximum_Age | Minimum_Age |
+----------------+-----------+-------------+-------------+-------------+
|              2 |        39 |     19.5000 |          20 |          19 |
+----------------+-----------+-------------+-------------+-------------+
1 row in set (0.028 sec)

mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.016 sec)

mysql>
```

9. SELECT with various clauses – GROUP BY, HAVING, ORDER BY

Output:

```
mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.015 sec)

mysql> SELECT Department_ID, COUNT(*) AS Total_Students
    -> FROM student
    -> GROUP BY Department_ID;
+---------------+----------------+
| Department_ID | Total_Students |
+---------------+----------------+
|             1 |              2 |
+---------------+----------------+
1 row in set (0.055 sec)

mysql> SELECT Department_ID, AVG(Age) AS Average_Age
    -> FROM student
    -> GROUP BY Department_ID;
+---------------+-------------+
| Department_ID | Average_Age |
+---------------+-------------+
|             1 |     19.5000 |
+---------------+-------------+
1 row in set (0.021 sec)

mysql> SELECT Department_ID, COUNT(*) AS Total_Students
    -> FROM student
    -> GROUP BY Department_ID
    -> HAVING COUNT(*) > 1;
+---------------+----------------+
| Department_ID | Total_Students |
+---------------+----------------+
```

```
    -> HAVING COUNT(*) > 1;
+---------------+----------------+
| Department_ID | Total_Students |
+---------------+----------------+
|             1 |              2 |
+---------------+----------------+
1 row in set (0.035 sec)

mysql> SELECT * FROM student
    -> ORDER BY Age ASC;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.062 sec)

mysql> SELECT * FROM student
    -> ORDER BY Age DESC;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.016 sec)

mysql> SELECT Department_ID, COUNT(*) AS Total_Students
    -> FROM student
    -> GROUP BY Department_ID
    -> HAVING COUNT(*) >= 1
    -> ORDER BY Total_Students DESC;
+---------------+----------------+
| Department_ID | Total_Students |
+---------------+----------------+
|             1 |              2 |
+---------------+----------------+
1 row in set (0.061 sec)
```

```
2 rows in set (0.016 sec)

mysql> SELECT Department_ID, COUNT(*) AS Total_Students
    -> FROM student
    -> GROUP BY Department_ID
    -> HAVING COUNT(*) >= 1
    -> ORDER BY Total_Students DESC;
+---------------+----------------+
| Department_ID | Total_Students |
+---------------+----------------+
|             1 |              2 |
+---------------+----------------+
1 row in set (0.061 sec)

mysql> SELECT City, COUNT(*) AS Total_Students
    -> FROM student
    -> GROUP BY City;
+---------+----------------+
| City    | Total_Students |
+---------+----------------+
| Chennai |              2 |
+---------+----------------+
1 row in set (0.062 sec)

mysql> SELECT * FROM student;
+------------+--------+-------------------+------+---------+---------------+
| Student_ID | Name   | Email             | Age  | City    | Department_ID |
+------------+--------+-------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+-------------------+------+---------+---------------+
2 rows in set (0.014 sec)
```

10. Query with SubQuery& Correlated Query

Output:

```
mysql>
mysql> USE dbms_lab;
Database changed
mysql> SELECT * FROM student;
+------------+--------+-------------------+------+---------+---------------+
| Student_ID | Name   | Email             | Age  | City    | Department_ID |
+------------+--------+-------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+-------------------+------+---------+---------------+
2 rows in set (0.014 sec)

mysql> SELECT * FROM student
    -> WHERE Age > (SELECT AVG(Age) FROM student);
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.089 sec)

mysql> SELECT Name FROM student
    -> WHERE Department_ID IN (
    ->     SELECT Department_ID
    ->     FROM department
    ->     WHERE Department_Name = 'CSE'
    -> );
+--------+
| Name   |
+--------+
| Hitesh |
| Rahul  |
+--------+
2 rows in set (0.411 sec)

mysql> SELECT * FROM student s
    -> WHERE Age > (
    ->     SELECT AVG(Age)
    ->     FROM student
    ->     WHERE Department_ID = s.Department_ID
```

```
    ->     SELECT AVG(Age)
    ->     FROM student
    ->     WHERE Department_ID = s.Department_ID
    -> );
+------------+-------+-----------------+------+---------+---------------+
| Student_ID | Name  | Email           | Age  | City    | Department_ID |
+------------+-------+-----------------+------+---------+---------------+
|        102 | Rahul | rahul@gmail.com |   20 | Chennai |             1 |
+------------+-------+-----------------+------+---------+---------------+
1 row in set (0.038 sec)

mysql> SELECT Name FROM student
    -> WHERE Department_ID IN (
    ->     SELECT Department_ID
    ->     FROM student
    ->     GROUP BY Department_ID
    ->     HAVING COUNT(*) > 5
    -> );
Empty set (0.041 sec)

mysql> SELECT * FROM department;
+---------------+-----------------+
| Department_ID | Department_Name |
+---------------+-----------------+
|             1 | CSE             |
|             2 | ECE             |
|             3 | MECH            |
+---------------+-----------------+
3 rows in set (0.020 sec)

mysql> SELECT * FROM student;
+------------+--------+------------------+------+---------+---------------+
| Student_ID | Name   | Email            | Age  | City    | Department_ID |
+------------+--------+------------------+------+---------+---------------+
|        101 | Hitesh | hitesh@gmail.com |   19 | Chennai |             1 |
|        102 | Rahul  | rahul@gmail.com  |   20 | Chennai |             1 |
+------------+--------+------------------+------+---------+---------------+
2 rows in set (0.016 sec)

mysql>
```